

Probability & Statistics

Final Exam

(MT2005)

Date: June 5th 2026

Course Instructor(s)

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Total Time (Hrs): 3

Total Marks: 100

Total Questions: 5

Roll No

Section

Student Signature

Do not write below this line

Attempt all the questions on Answer Book.

Statistical tables and a formula sheet are provided. Personal tables/formula sheets are not permitted.

Lead Pencil work wouldn't be marked or claimed for rechecking. Use permanent ink pen.

Ensure the final answer is expressed in decimal form to **4 significant figures (not as a ratio)**.

If the data is split or divided, make sure to **display the resulting (new) dataset/variable** clearly.

If data is ambiguous, make an assumption and proceed—do not ask the invigilator.

Please read carefully and **choose the correct variables** and units before attempting your solution.

If you select the **wrong variable(s), incorrect values and wrong cases** etc., your entire answer will be awarded **zero marks**, even if the working steps are shown.

DO NOT ATTACH QUESTION PAPER.

CLO 1: Statistical Data Interpretation: Analyze and interpret various data types by computing measures of central tendency and dispersion, constructing frequency distributions, and utilizing graphical techniques for precise data representation. [wt:10]

Q1:

Marks[2+25+3=30]

Statistical Analysis of Cloud Backend System Performance

A cloud services company is evaluating the performance of two backend systems, System A and System B, under identical load conditions. The monitoring tool records the following for each request:

- Start Time (in milliseconds)
- End Time (in milliseconds)
- Data Size Transferred (in KB)

The company does not directly store response time. Instead, it is computed as:

$$\text{Response Time} = \text{End Time} - \text{Start Time}$$

Management suspects inconsistencies in system performance and requires a statistical analysis to compare both systems in terms of speed and stability.

System A Logs

Request	Start Time	End Time	Data Size (KB)
1	102	118	50
2	110	130	55
3	115	140	60
4	120	150	65
5	125	145	55
6	130	155	70
7	135	160	75
8	140	165	80
9	145	175	85
10	150	190	90

System B Logs

Request	Start Time	End Time	Data Size (KB)
1	200	210	40
2	205	220	45
3	210	230	50
4	215	240	55
5	220	250	60
6	225	255	65
7	230	260	70
8	235	270	75
9	240	275	80
10	245	310	85

- Using the given data, compute the response time for each request for both System A and B.
- For both systems, compute the five-number summary for response time. Construct comparative box plots on the same scale for response time (System A vs System B) and identify any outliers. Explain whether the presence of outliers indicates occasional performance bottlenecks or unstable system behavior. Also comment on whether the distribution appears symmetric or skewed for each system?
- Compute the Interquartile Range (IQR) for response time for both System A and System B. Using these values, determine which system shows more consistent response time performance across requests. Justify your conclusion based on the spread of the data.

CLO 2: Probability and Distribution Analysis: Apply foundational principles of probability to analyze experiments, including Bayes' theorem, evaluate discrete and continuous distributions, and explore applications in machine learning. [wt:15]

Q2:

Marks[5+5+2+2+1=15]

Capacity Planning in Software Engineering

A software engineering company in Lahore is developing a cloud-based analytics system to handle increasing user demand. The project is structured into two successive phases: *Stage 1: System Design* and *Stage 2: Develop and Deploy*. The time for each stage is random because of uncertainty in system requirements and debugging complexity. Let X be random variable representing the completion time (in months) for Design stage and Y represents the completion time (in months) for Development stage. Based on Past projects, completion time are 2, 3, and 4 months for design stage and 6, 7 and 8 months for development stage. Each Possible time for X and Y is equally likely, and X and Y are independent variables. Management has set the deadline of 10 months for the completion of entire project.

- Construct a joint probability distribution of X and Y
- Find the probability distribution of T from the joint distribution, where T is the total time required to complete the project i.e. $T=X+Y$.
- Determine the expected total completion time.

- d) Quantify the dispersion in total completion time.
- e) Calculate the probability that the project will be completed within 10 months.

Q3:

Marks[10+10=20]

- a) A technology company is launching a new software product in a competitive digital market. Product success is measured by its market share, which is influenced by user adoption, system performance and marketing efficiency. If the product achieves a market share of 10% or more in the next two years, the product will be called a “success”. The marketing department has considered all possibilities and decided that it expects the product to attain a market share of 12% this time. However, this estimate is not certain. The standard deviation is estimated to be 3%, indicating the uncertainty in the estimate. Market share is assumed to follow a normal distribution.
 - a) Find the probability that the new project will be successful.
 - b) If the company wants to be even more selective and define “extraordinary success” as top 5%, find the corresponding market share threshold.
- b) The number of packet arrivals at a high-speed router used in a cloud-data center is analyzed by a network engineer. The engineering team is more concerned with traffic instability (burstiness). From historical measurements, it is observed that the variance of the packet arrivals per minute is $1.8 \text{ Packets}^2 \text{ per minutes}^2$
 - a) The router is in a congestion state if it receives more than 2 packets in a minute. Compute the probability of congestion.
 - b) The system is upgraded to handle peak traffic safely if it can support loads up to 3 packets in a 15-second interval. Find the probability that the actual traffic demand exceeds this capacity.

CLO 3: Inferential and Predictive Analysis: Estimate parameters, perform hypothesis testing, and employ regression analysis to model relationships between variables, assess coefficient significance, and confidently predict future outcomes. **[wt:25]**

Q4:

[Marks=10]

Correct credentials may sometimes be used to gain unauthorized access to computer systems. To fight this, cybersecurity systems will examine a user's typing behavior, such as, the time between keystrokes, to identify legitimate users from potential intruders. For this purpose, the cybersecurity team analyzes the keystroke pattern of legitimate users. Based on a sample of 40 users and assuming a known population standard deviation, the team reports that it is 95% confident that true average time between keystrokes for legitimate users lies between 0.30 and 0.50 seconds, which represents typical typing behavior of authorized users. Later, for stricter security, the team considers decreasing the significance level from 5% to 1%. Determine the revised confidence interval for true average time between keystrokes by considering this new level significance.

Q5:

Marks[9+16=25]

A software engineer is investigating how model complexity influences the inference time of an AI system. Model complexity is measured on a scale from 1 to 8. During testing, the following observations were made:

At complexity level 1, the inference time was 12 ms, increasing to 15 ms at level 2. When the complexity reached 3, two experimental runs produced 18 ms and 17 ms, indicating slight variation. At complexity level 4, the recorded inference time was 20 ms. Increasing the complexity to 5 resulted in an inference time of 23 ms, although an optimized execution under the same conditions recorded 21 ms. Further increases to levels 6, 7, and 8 resulted in inference times of 25 ms, 27 ms, and 30 ms, respectively. After performing a preliminary analysis, the engineer observed that: when model complexity is hypothetically zero, the expected inference time is approximately 9.82 ms, and for every unit increase in model complexity, inference time increases by about 2.4955 ms. The engineer now wishes to further validate and interpret this relationship.

- a) Assess the extent to which variations in inference time can be explained by changes in model complexity, and comment on the strength of the relationship obtained.
- b) Partition the total variation in inference time into its relevant components. Based on the partition, use ANOVA (with proper labeling) to determine whether there is sufficient statistical evidence at the 5% level of significance to conclude that the overall regression model is statistically significant. Justify your conclusion using the appropriate test statistic and its corresponding p-value (0.0001).

FORMULA SHEET FINAL EXAM SPRING 2026

$$\bar{x} \pm Z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

$$z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$$

$$\bar{x} \pm t_{(\alpha/2, v)} \frac{s}{\sqrt{n}}$$

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}}$$

$$b_1 = \frac{\sum(x - \bar{x})(y - \bar{y})}{\sum(x - \bar{x})^2} = \frac{n \sum xy - \sum x \sum y}{n \sum x^2 - (\sum x)^2}$$

$$b_0 = \bar{y} - b_1 \bar{x}$$

$$r = \frac{\sum(x - \bar{x})(y - \bar{y})}{\sqrt{\sum(x - \bar{x})^2 \sum(y - \bar{y})^2}}$$

$$r = \frac{n \sum xy - \sum x \sum y}{\sqrt{(n \sum x^2 - (\sum x)^2)(n \sum y^2 - (\sum y)^2)}}$$

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}}$$

$$t = \frac{b_1 - \beta_1}{s_{yx}/\sqrt{S_{xx}}} \text{ or } \frac{b_1 - \beta_1}{s_{b1}}$$

$$s_{yx} = \sqrt{\frac{\sum(Y - \hat{y})^2}{n-2}}$$

$$S_{xx} = \sum(X - \bar{X})^2$$

$$\text{Var}(X) = E(X^2) - [E(X)]^2$$

$$s_x = \sqrt{\frac{\sum(X - \bar{X})^2}{n-1}}, \sigma_x = \sqrt{\frac{\sum(X - \mu)^2}{N}}$$

$$P(A \cap B \cap C) = P(A)P(B)P(C)$$

$$P(A \cap B \cap C) = P(A)P(B/A)P(C/A \cap B)$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$F = \frac{MSR}{MSE}$$

$$P(B_r/A) = \frac{P(B_r \cap A)}{\sum_i^k P(B_i \cap A)} = \frac{P(B_r)P(A/B_r)}{\sum_i^k P(B_i)P(A/B_i)}, r = 1, 2, \dots, k$$

$$b(x; n, p) = \binom{n}{x} p^x q^{n-x}, x = 0, 1, \dots, n, q = 1 - p,$$

$$f(x_1, x_2, \dots, x_k; p_1, p_2, \dots, p_k; n) = \binom{n}{x_1, x_2, \dots, x_k} p_1^{x_1} p_2^{x_2} \dots p_k^{x_k}$$

$$h(x; N, n, k) = \frac{\binom{k}{x} \binom{N-k}{n-x}}{\binom{N}{n}}$$

$$p(x; \lambda t) = \frac{e^{-\lambda t} (\lambda t)^x}{x!} \text{ or } p(x; \mu) = \frac{e^{-\mu} (\mu)^x}{x!}, x = 0, 1, 2, \dots,$$

$$R^2 = 1 - \frac{SSE}{SST} \text{ or } \frac{SSR}{SST}$$

$$SSR = \sum(\hat{y} - \bar{Y})^2$$

$$SST = \sum(Y - \bar{Y})^2 = \sum Y^2 - \frac{(\sum Y)^2}{n}$$

$$\sum e_i^2 = SSE = \sum(Y - \hat{y})^2 = \sum Y^2 - b_0 \sum y - b_1 \sum XY$$

$$\mu = E(X)$$

$$E(X) = \sum x f(x) \quad E(X^2) = \sum x^2 f(x)$$

$$t = \frac{\bar{d} - d_0}{s_d/\sqrt{n}}$$

$$\bar{d} \pm t_{(\alpha/2, v)} s_d/\sqrt{n}$$

$$z = \frac{x - \mu}{\sigma}$$

$$\text{Lower limit} = Q_1 - 1.5(IQR)$$

$$\text{Upper limit} = Q_3 + 1.5(IQR)$$

National University of Computer and Emerging Sciences

Lahore Campus

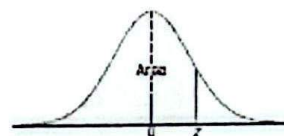


Table A.3 Areas under the Normal Curve

<i>z</i>	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
-3.4	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0002
-3.3	0.0005	0.0005	0.0005	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0003
-3.2	0.0007	0.0007	0.0006	0.0006	0.0006	0.0006	0.0006	0.0005	0.0005	0.0005
-3.1	0.0010	0.0009	0.0009	0.0009	0.0008	0.0008	0.0008	0.0008	0.0007	0.0007
-3.0	0.0013	0.0013	0.0013	0.0012	0.0012	0.0011	0.0011	0.0011	0.0010	0.0010
-2.9	0.0019	0.0018	0.0018	0.0017	0.0016	0.0016	0.0015	0.0015	0.0014	0.0014
-2.8	0.0026	0.0025	0.0024	0.0023	0.0023	0.0022	0.0021	0.0021	0.0020	0.0019
-2.7	0.0035	0.0034	0.0033	0.0032	0.0031	0.0030	0.0029	0.0028	0.0027	0.0026
-2.6	0.0047	0.0045	0.0044	0.0043	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036
-2.5	0.0062	0.0060	0.0059	0.0057	0.0055	0.0054	0.0052	0.0051	0.0049	0.0048
-2.4	0.0082	0.0080	0.0078	0.0075	0.0073	0.0071	0.0069	0.0068	0.0066	0.0064
-2.3	0.0107	0.0104	0.0102	0.0099	0.0096	0.0094	0.0091	0.0089	0.0087	0.0084
-2.2	0.0139	0.0136	0.0132	0.0129	0.0125	0.0122	0.0119	0.0116	0.0113	0.0110
-2.1	0.0179	0.0174	0.0170	0.0166	0.0162	0.0158	0.0154	0.0150	0.0146	0.0143
-2.0	0.0228	0.0222	0.0217	0.0212	0.0207	0.0202	0.0197	0.0192	0.0188	0.0183
-1.9	0.0287	0.0281	0.0274	0.0268	0.0262	0.0256	0.0250	0.0244	0.0239	0.0233
-1.8	0.0359	0.0351	0.0344	0.0336	0.0329	0.0322	0.0314	0.0307	0.0301	0.0294
-1.7	0.0446	0.0436	0.0427	0.0418	0.0409	0.0401	0.0392	0.0384	0.0375	0.0367
-1.6	0.0548	0.0537	0.0526	0.0516	0.0505	0.0495	0.0485	0.0475	0.0465	0.0455
-1.5	0.0668	0.0655	0.0643	0.0630	0.0618	0.0606	0.0594	0.0582	0.0571	0.0559
-1.4	0.0808	0.0793	0.0778	0.0764	0.0749	0.0735	0.0721	0.0708	0.0694	0.0681
-1.3	0.0968	0.0951	0.0934	0.0918	0.0901	0.0885	0.0869	0.0853	0.0838	0.0823
-1.2	0.1151	0.1131	0.1112	0.1093	0.1075	0.1056	0.1038	0.1020	0.1003	0.0985
-1.1	0.1357	0.1335	0.1314	0.1292	0.1271	0.1251	0.1230	0.1210	0.1190	0.1170
-1.0	0.1587	0.1562	0.1539	0.1515	0.1492	0.1469	0.1446	0.1423	0.1401	0.1379
-0.9	0.1841	0.1814	0.1788	0.1762	0.1736	0.1711	0.1685	0.1660	0.1635	0.1611
-0.8	0.2119	0.2090	0.2061	0.2033	0.2005	0.1977	0.1949	0.1922	0.1894	0.1867
-0.7	0.2420	0.2389	0.2358	0.2327	0.2296	0.2266	0.2236	0.2206	0.2177	0.2148
-0.6	0.2743	0.2709	0.2676	0.2643	0.2611	0.2578	0.2546	0.2514	0.2483	0.2451
-0.5	0.3085	0.3050	0.3015	0.2981	0.2946	0.2912	0.2877	0.2843	0.2810	0.2776
-0.4	0.3446	0.3409	0.3372	0.3336	0.3300	0.3264	0.3228	0.3192	0.3156	0.3121
-0.3	0.3821	0.3783	0.3745	0.3707	0.3669	0.3632	0.3594	0.3557	0.3520	0.3483
-0.2	0.4207	0.4168	0.4129	0.4090	0.4052	0.4013	0.3974	0.3936	0.3897	0.3859
-0.1	0.4602	0.4562	0.4522	0.4483	0.4443	0.4404	0.4364	0.4325	0.4286	0.4247
-0.0	0.5000	0.4960	0.4920	0.4880	0.4840	0.4801	0.4761	0.4721	0.4681	0.4641

Table A.3 (continued) Areas under the Normal Curve

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997
3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998

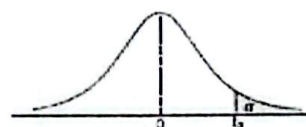


Table A.4 Critical Values of the t-Distribution

v	α						
	0.40	0.30	0.20	0.15	0.10	0.05	0.025
1	0.325	0.727	1.376	1.963	3.078	6.314	12.706
2	0.289	0.617	1.061	1.386	1.886	2.920	4.303
3	0.277	0.584	0.978	1.250	1.638	2.353	3.182
4	0.271	0.569	0.941	1.190	1.533	2.132	2.776
5	0.267	0.559	0.920	1.156	1.476	2.015	2.571
6	0.265	0.553	0.906	1.134	1.440	1.943	2.447
7	0.263	0.549	0.896	1.119	1.415	1.895	2.365
8	0.262	0.546	0.889	1.108	1.397	1.860	2.306
9	0.261	0.543	0.883	1.100	1.383	1.833	2.262
10	0.260	0.542	0.879	1.093	1.372	1.812	2.228
11	0.260	0.540	0.876	1.088	1.363	1.796	2.201
12	0.259	0.539	0.873	1.083	1.356	1.782	2.179
13	0.259	0.538	0.870	1.079	1.350	1.771	2.160
14	0.258	0.537	0.868	1.076	1.345	1.761	2.145
15	0.258	0.536	0.866	1.074	1.341	1.753	2.131
16	0.258	0.535	0.865	1.071	1.337	1.746	2.120
17	0.257	0.534	0.863	1.069	1.333	1.740	2.110
18	0.257	0.534	0.862	1.067	1.330	1.734	2.101
19	0.257	0.533	0.861	1.066	1.328	1.729	2.093
20	0.257	0.533	0.860	1.064	1.325	1.725	2.086
21	0.257	0.532	0.859	1.063	1.323	1.721	2.080
22	0.256	0.532	0.858	1.061	1.321	1.717	2.074
23	0.256	0.532	0.858	1.060	1.319	1.714	2.069
24	0.256	0.531	0.857	1.059	1.318	1.711	2.064
25	0.256	0.531	0.856	1.058	1.316	1.708	2.060
26	0.256	0.531	0.856	1.058	1.315	1.706	2.056
27	0.256	0.531	0.855	1.057	1.314	1.703	2.052
28	0.256	0.530	0.855	1.056	1.313	1.701	2.048
29	0.256	0.530	0.854	1.055	1.311	1.699	2.045
30	0.256	0.530	0.854	1.055	1.310	1.697	2.042
40	0.255	0.529	0.851	1.050	1.303	1.684	2.021
60	0.254	0.527	0.848	1.045	1.296	1.671	2.000
120	0.254	0.526	0.845	1.041	1.289	1.658	1.980
∞	0.253	0.524	0.842	1.036	1.282	1.645	1.960

Table A.4 (continued) Critical Values of the *t*-Distribution

<i>v</i>	α						
	0.02	0.015	0.01	0.0075	0.005	0.0025	0.0005
1	15.894	21.205	31.821	42.433	63.656	127.321	636.578
2	4.849	5.643	6.965	8.073	9.925	14.089	31.600
3	3.482	3.896	4.541	5.047	5.841	7.453	12.924
4	2.999	3.298	3.747	4.088	4.604	5.598	8.610
5	2.757	3.003	3.365	3.634	4.032	4.773	6.869
6	2.612	2.829	3.143	3.372	3.707	4.317	5.959
7	2.517	2.715	2.998	3.203	3.499	4.029	5.408
8	2.449	2.634	2.896	3.085	3.355	3.833	5.041
9	2.398	2.574	2.821	2.998	3.250	3.690	4.781
10	2.359	2.527	2.764	2.932	3.169	3.581	4.587
11	2.328	2.491	2.718	2.879	3.106	3.497	4.437
12	2.303	2.461	2.681	2.836	3.055	3.428	4.318
13	2.282	2.436	2.650	2.801	3.012	3.372	4.221
14	2.264	2.415	2.624	2.771	2.977	3.326	4.140
15	2.249	2.397	2.602	2.746	2.947	3.286	4.073
16	2.235	2.382	2.583	2.724	2.921	3.252	4.015
17	2.224	2.368	2.567	2.706	2.898	3.222	3.965
18	2.214	2.356	2.552	2.689	2.878	3.197	3.922
19	2.205	2.346	2.539	2.674	2.861	3.174	3.883
20	2.197	2.336	2.528	2.661	2.845	3.153	3.850
21	2.189	2.328	2.518	2.649	2.831	3.135	3.819
22	2.183	2.320	2.508	2.639	2.819	3.119	3.792
23	2.177	2.313	2.500	2.629	2.807	3.104	3.768
24	2.172	2.307	2.492	2.620	2.797	3.091	3.745
25	2.167	2.301	2.485	2.612	2.787	3.078	3.725
26	2.162	2.296	2.479	2.605	2.779	3.067	3.707
27	2.158	2.291	2.473	2.598	2.771	3.057	3.689
28	2.154	2.286	2.467	2.592	2.763	3.047	3.674
29	2.150	2.282	2.462	2.586	2.756	3.038	3.660
30	2.147	2.278	2.457	2.581	2.750	3.030	3.646
40	2.123	2.250	2.423	2.542	2.704	2.971	3.551
60	2.099	2.223	2.390	2.504	2.660	2.915	3.460
120	2.076	2.196	2.358	2.468	2.617	2.860	3.373
∞	2.054	2.170	2.326	2.432	2.576	2.807	3.290